



Navigating the queue: A dynamic pricing framework for ride-sourcing platform with heterogeneous demand

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ABSTRACT

Ride-sourcing platforms commonly employ dynamic pricing to regulate travel demand and dispatch or relocate vehicles to manage supply distribution. However, existing strategies often overlook the heterogeneity in spatial-temporal demand distribution and diverse customer behaviors. To address this gap, this paper proposes an origin-destination (OD)-based dynamic pricing framework that integrates link-node-level vehicle dispatching and relocation decisions, while accounting for heterogeneous demand and customer queue-jumping behavior. In this framework, customers, characterized by varying values of time (VOT) and willingness to pay (WTP), can choose to wait, abandon the trip, or pay a premium to jump the queue. To model this heterogeneous behavior, we introduce an integral-form demand function that captures elasticity with respect to both VOT and WTP. The dynamic pricing problem is formulated as a constrained optimization problem that jointly determines OD-based pricing and queue-jumping premiums. A rolling horizon approach, featuring a short decision interval with a longer planning horizon, is introduced to capture queueing dynamics that extend across periods, thereby supporting both real-time responsiveness and long-term optimization. An event-triggered control scheme is further incorporated to reduce computational complexity and stabilize pricing fluctuations. An Approximate Dynamic Programming (ADP) approach is adopted to manage vehicle dispatching and relocation at the link-node level. Empirical validation on the Manhattan road network confirmed the effectiveness of the proposed approach in improving platform profitability, increasing demand satisfaction, and enhancing overall system efficiency.

1. Introduction

Ride-sourcing platforms, such as Didi, Uber, and Lyft, which use Internet-based technology to match passengers and drivers, have revolutionized the transportation industry and the form of personal mobility (Wang and Yang, 2019; Qin et al., 2022). When customers request a ride, platforms quote a trip fare and an estimated waiting time based on the spatial-temporal distribution of

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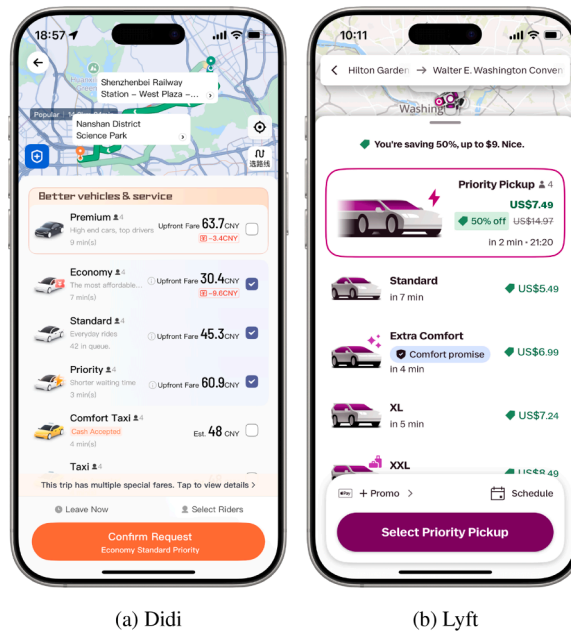


Fig. 1. Screenshots of Didi and Lyft apps.

demand and supply. During periods of high customer demand, platforms may apply surge pricing to the base fare and quote longer estimated waiting times. Customers can then decide whether to submit or cancel their requests.

Once customers confirm their ride requests, they typically join a matching queue to await driver assignment. Many platforms also allow customers to obtain higher matching priority by paying an additional premium. This practice has been widely adopted in real-world ride-sourcing platforms. For example, as illustrated in Fig. 1, Didi offers multiple service categories, including Economy, Standard, and Priority, with different fare levels and expected waiting times. Similarly, Lyft offers Standard and Priority Pickup services, where customers who pay a higher fare are promised shorter pickup times. Although these services are marketed as differentiated products, they essentially implement priority-based matching schemes in which higher-paying customers receive preferential access to available vehicles. Consequently, customers with higher values of time (VOT) and willingness to pay (WTP) are generally more inclined to choose surge pricing or premium services in exchange for shorter waiting times. This practical observation provides strong practical motivation for explicitly modeling queue-jumping behavior in ride-sourcing systems.

Dynamic pricing has long been recognized as an effective approach for demand regulation in ride-sourcing systems. Early studies primarily focused on static pricing, assuming comprehensive knowledge of the system to determine optimal pricing policies under equilibrium conditions (Wang et al., 2016; Fang et al., 2019; Banerjee et al., 2022). To better cope with the uncertainty and non-stationarity of real-world environments, subsequent research shifted toward dynamic pricing, which has been shown to be more robust to fluctuations in system parameters (Banerjee et al., 2015) and more effective in promoting spatial-temporal equilibrium. Dynamic pricing has also been integrated with various incentive mechanisms, such as reward schemes for customers (Yang et al., 2020) and wage optimization for drivers (Bai et al., 2019), to further improve market efficiency. Despite these advances, existing pricing models generally rely on aggregate demand representations and pay limited attention to customer heterogeneity. Moreover, the priority service tiers offered by ride-sourcing platforms provide strong practical motivation for explicitly modeling queue-jumping behavior, yet such mechanisms remain largely unexplored in the literature.

In addition to demand-side pricing, ride-sourcing platforms can alleviate supply-demand imbalances through vehicle dispatching and relocation, which play a critical role in improving fleet utilization and enhancing operational efficiency (Zhu et al., 2024; Guo et al., 2026). To address these large-scale sequential decision-making problems, approximate dynamic programming (ADP) and reinforcement learning (RL) have emerged as effective solution approaches for high-dimensional and stochastic ride-sourcing systems (Ke et al., 2022; Huang et al., 2023, 2025). Building on these methodologies, recent studies have investigated the joint optimization of pricing and vehicle operations by integrating dispatching and relocation decisions to improve overall system performance (Chen et al., 2019; Lei et al., 2020; Liang et al., 2021; Zhang et al., 2024; Sun et al., 2024; Zhang et al., 2025). However, most existing studies optimize pricing and vehicle operations at the same spatial and temporal resolutions, making it difficult to simultaneously exploit OD-level demand information and link-node-level operational flexibility.

Another practical challenge of dynamic pricing lies in its tendency to introduce undesirable price volatility in response to short-term fluctuations in supply and demand (Yan et al., 2020). Rolling horizon optimization provides an effective approach for addressing such sequential decision-making problems by repeatedly updating decisions as new information becomes available. It has been successfully applied to traffic state prediction (Pan et al., 2013; Zhong et al., 2017), trajectory planning (Xu and Geroliminis, 2026), taxi

dispatching (Miao et al., 2015), car-sharing (Li et al., 2025) and logistics (Tran and Stokkink, 2026) systems. These studies demonstrate its ability to balance real-time responsiveness with long-term decision quality, making it a promising approach for developing stable and adaptive dynamic pricing strategies in ride-sourcing systems.

To address these challenges, this paper proposes an integrated OD-based dynamic pricing framework that incorporates customer heterogeneity and queue-jumping behavior. Customers make pricing and queue-jumping decisions according to their individual VOT and WTP, while unmatched requests remain in the system until they are either matched or time out. A rolling horizon framework is adopted to capture the inter-temporal dynamics of the matching queue, and an event-triggered pricing mechanism is introduced to reduce unnecessary price fluctuations and computational burden. Meanwhile, vehicle dispatching and relocation are optimized at the link-node level using ADP, enabling the coordinated optimization of demand regulation and supply management across multiple spatial scales. The main contributions of this paper are summarized as follows:

- Propose an OD-based dynamic pricing framework that integrates link-node-level vehicle dispatching and relocation decisions to address heterogeneous spatial-temporal demand distribution.
- Model customer heterogeneity in VOT and WTP, and incorporate a queue-jumping premium mechanism to reflect individual waiting and queue-jumping decisions.
- Adopt a rolling horizon approach to capture queueing demand that carries over into subsequent periods, combined with an event-triggered scheme to reduce abrupt price fluctuations.
- Apply ADP to manage detailed vehicle dispatching and relocation decisions at the link-node level efficiently.

The remainder of this paper is organized as follows. Section 2 reviews the existing literature on customer participation and heterogeneity, demand-side modeling and pricing strategies, and joint pricing and vehicle operations in ride-sourcing systems. Section 3 presents the OD-based dynamic pricing scheme incorporating a queue-jumping mechanism and models elastic customer demand based on individuals' VOT and WTP. The dynamic pricing problem is formulated under a rolling horizon framework and coupled with an event-triggered scheme, while the vehicle dispatching and relocation problem at the link-node level is presented, along with an ADP solution. Section 4 evaluates the performance of the proposed framework under different demand patterns and demonstrates the benefits of incorporating queue-jumping behavior, the rolling horizon approach, and the event-triggered scheme. The proposed ADP approach is also compared against several baseline methods, while a sensitivity analysis is conducted to examine the robustness of the framework under zone partitioning and driver compliance rates. Finally, Section 5 concludes the study and outlines potential directions for future research.

2. Related work

2.1. Customer participation and heterogeneity

Customer participation and customer heterogeneity have received increasing attention in ride-sourcing pricing studies. Yang et al. (2020) integrated a reward scheme with surge pricing, where each customer is assumed to have a balance account. Customers pay an additional fee during peak periods and use the accumulated balance to subsidize trips during off-peak periods. However, customers do not explicitly decide whether to participate in the additional payment mechanism, despite the fact that participation is closely related to platform service quality (Stokkink et al., 2025). In contrast, auction-based approaches have been studied as a way to incorporate customer participation in pricing and service allocation (Karamanis et al., 2020; Buchholz et al., 2025). These mechanisms provide customers with explicit interaction through bidding, allowing them to influence service outcomes. However, these approaches rely on strategic bidding and active customer participation, increasing implementation complexity and limiting their practicality in real-time ride-sourcing systems. Moreover, bidding-based mechanisms may reduce pricing transparency because service outcomes depend on competing bids rather than posted prices.

Customers inherently exhibit substantial heterogeneity in their VOT and WTP. Effective pricing strategies should price rides according to complex underlying demand functions with heterogeneous customers and detect non-stationary behavior (Abkarian and Mahmassani, 2024). Zhang et al. (2018) evaluated different regulatory policies for a monopoly public transit service with passengers having heterogeneous VOT and WTP. Wu et al. (2025) introduced a heterogeneous VOT distribution to design optimal pricing for passenger choices between solo and pooled rides. In the context of ride-sourcing, when customers have the option to pay queue-jumping premiums based on their VOT and WTP, two key issues remain understudied: how to model such queue-jumping behavior, which is often observed in real-world scenarios, and how to design dynamic pricing and premiums to accommodate heterogeneous customer behaviors.

2.2. Demand-side modeling and pricing strategies

Static pricing, dynamic pricing, and subsidy strategies have been widely studied as demand regulation mechanisms in ride-sourcing systems. Wang et al. (2016) modeled the ride-sourcing market using an aggregate static pricing approach and examined the existence and stability of market equilibria. Subsequent studies further investigated static pricing under different optimization objectives, including maximizing platform revenue, social welfare, and throughput (Fang et al., 2019; Banerjee et al., 2022). However, static pricing requires comprehensive knowledge of the system and cannot effectively capture the spatial-temporal dynamics of demand and supply. In contrast, dynamic pricing has been shown to offer significant advantages over static pricing, particularly in nonstationary

environments (Chen et al., 2024). It is more robust to fluctuations in system parameters (Banerjee et al., 2015) and effective in promoting spatial-temporal equilibrium. A representative example is surge pricing during peak periods, which serves to reallocate supply in response to rising demand (Cohen et al., 2016). Moreover, Castillo et al. (2017) highlighted the role of surge pricing in mitigating the “Wild Goose Chase” (WGC) phenomenon, in which drivers are matched with distant passengers, resulting in longer passenger waiting times and lower driver availability.

In the context of dynamic pricing, Zha et al. (2018) developed a spatial pricing model based on the intra-zone matching assumption, which enables zone-specific pricing but falls short in capturing the spatial heterogeneity of supply and demand across different regions. Tang et al. (2022) subsequently investigated optimal spatial pricing using a bi-level optimization framework, where the platform determines spatially differentiated prices in the upper level while solving trip distribution among zones in the lower level. To account for spatial imbalances and temporal variations in supply and demand, Yan et al. (2020) introduced a dynamic waiting pool-matching mechanism integrated with dynamic pricing to reduce price volatility and improve social welfare. Ma et al. (2022) proposed a spatial-temporal pricing mechanism that achieves incentive alignment by making driver trip acceptance a subgame-perfect equilibrium strategy, thereby improving social welfare and earning equity. Xu et al. (2022) examined dynamic pricing strategies with a time-dependent matching model that explicitly incorporates order cancellation behavior. Dynamic pricing has also been integrated with various incentive mechanisms, such as customer reward schemes (Yang et al., 2020) and driver wage optimization (Bai et al., 2019). To capture the impact of dynamic pricing on travel demand, existing studies typically model demand as an aggregate elastic function of travel cost, for example, using the Inverse Product Differentiation Logit model (Gómez-Lobo et al., 2022). However, such formulations generally overlook individual customer heterogeneity in the decision-making process. Moreover, they typically assume that demand is either immediately matched or permanently lost within each decision period, ignoring customers who are willing to remain in the matching queue. In practice, customers may choose to wait based on their individual trade-offs between travel time and cost (Wu et al., 2025). Modeling these inter-temporal queueing dynamics is essential for accurately characterizing customer behavior under dynamic pricing, yet remains largely unexplored in the existing literature.

2.3. Joint pricing and vehicle operations

For the two-sided ride-sourcing market, the platform can alleviate supply-demand disequilibrium via vehicle dispatching and relocation. Lei et al. (2020) proposed a bi-level mathematical program with equilibrium constraints, optimizing path-based pricing

Table 1
Nomenclature.

Variable	Description
System setting	
$\mathbb{T}$	Simulation horizon set: $t \in \mathbb{T} = \{0, 1, 2, \dots, T-1\}$
T	Number of periods in the simulation horizon
δt	Decision interval
t_p	Planning horizon
Z	Zone set: $i, j \in Z$
N	Node set: $l \in N$
E	Edge set
Demand side	
C	Customer set: $c \in C$
$\gamma_{i,j,t}$	Price multiplier for OD(i, j) at time t
$b_{i,j,t}$	Base fare for OD(i, j) at time t
$p_{i,j,t}$	Trip fare for OD(i, j) at time t
$u_{i,j,t}$	Queue-jumping premium for OD(i, j) at time t
$h_{i,j,t}$	Estimated waiting time for OD(i, j) at time t
$d_{i,j,t}$	Reserved demand rate for OD(i, j) at time t
$g_{i,j,t}^t$	Total cost including trip fare and queue-jumping premium for OD(i, j) at time t
g_c^w	Generalized cost including trip fare and waiting time cost for customer c
$n_{i,j,t}$	Total number of customers with OD(i, j) at time t
$\hat{n}_{i,j,t}$	Number of newly arriving customers with OD(i, j) at time t
v_c	Customer's value of time (VOT)
w_c	Customer's willingness to pay (WTP)
Supply side	
Q	Vehicle set: $q \in Q$
e	Travel cost per unit time
$m_{i,t}$	Number of idle vehicles in zone i at time t
$\tau_{o,d}$	Base travel time from origin o to destination d
$\hat{\tau}_{o,d,t}$	Estimated travel time from origin o to destination d at departure time t
$L_{o,d}$	Link-based distance between origin o and destination d
$x_{q,c,t}$	Dispatching status of vehicle q with customer c at time t
$y_{q,l,t}$	Relocation status of vehicle q relocated to node l at time t

and vehicle dispatching in the upper level while modeling travelers' equilibrium path choices in the lower level, and solved the problem using ADP. [Chen et al. \(2019\)](#) simultaneously optimized pricing and dispatch strategies using contextual bandits and temporal difference learning, while [Zhang et al. \(2024\)](#) further extended this framework by incorporating vehicle repositioning and solving the problem using contextual bandits and multi-agent deep reinforcement learning. [Sun et al. \(2024\)](#) formulated the interactions among vehicle dispatching, service pricing, and customer demand as a two-stage Stackelberg game under both independent and competitive pricing schemes. [Zhang et al. \(2025\)](#) further integrated re-dispatching and pricing strategies to optimize short-term revenue while strategically determining fleet size and initial vehicle allocation to enhance long-term system welfare. These studies demonstrate the benefits of jointly optimizing pricing and vehicle operations for improving system performance. However, they generally focus on integrated optimization frameworks without explicitly investigating how the spatial granularity of pricing and vehicle operations affects system performance.

To further improve system performance, recent studies have investigated the spatial granularity of pricing and vehicle operations. [Bimpikis et al. \(2019\)](#) proposed a time-invariant equilibrium model to examine spatial price discrimination. [Shi et al. \(2022\)](#) designed a multi-region dynamic pricing algorithm to set differentiated unit prices for different non-overlapping regions. [Lei and Ukkusuri \(2023\)](#) developed an enhanced twin-delayed deep determinant policy gradient (TD3) RL algorithm for pricing zones ranging from small grid networks to the New York City (NYC) network. Notably, the majority of zone-based pricing schemes adopt an origin-based pricing strategy, where prices are determined solely by the customer's departure zone. While origin-based pricing ensures consistency for trips from the same location, it fails to account for trip distance and destination attractiveness, resulting in homogeneous pricing across different origin-destination (OD) pairs and limiting its effectiveness in demand management. In contrast, OD-based pricing enables differentiated fare regulation based on both trip origins and destinations, thereby better capturing OD-specific demand and facilitating long-term supply balancing. Meanwhile, existing studies commonly perform driver-customer matching at the regional scale ([Beojone et al., 2024](#)), whereas link-node-level modeling provides much finer spatial granularity for real-time routing and dispatching ([Garg and Ranu, 2018](#); [Zhou et al., 2020](#); [Yu and Shen, 2020](#); [Liang et al., 2021](#)). To bridge this spatial mismatch, [Huang et al. \(2024\)](#) proposed a bi-level framework integrating link-node-level vehicle route guidance with regional vehicle relocation. These studies demonstrate the effectiveness of jointly optimizing pricing and vehicle operations for improving ride-sourcing system performance. However, most existing approaches either optimize pricing and vehicle operations at the same spatial scale or perform vehicle matching at a coarse regional level, making it difficult to simultaneously exploit OD-level demand information and fine-grained link-node-level operational flexibility.

3. Problem description

To facilitate readers' understanding, [Table 1](#) presents some critical variable notations used throughout the paper.

3.1. Basic settings

To better illustrate the proposed framework, [Fig. 2](#) compares the conventional ride-sourcing framework with the proposed one. In the conventional framework ([Fig. 2a](#)), customers submit trip requests to the platform and receive fare quotes and estimated waiting times, based on which they decide whether to accept and join the matching queue. The platform continuously monitors vehicle availability and subsequently dispatches vehicles to matched customers while relocating idle vehicles to balance supply and demand. Pricing, dispatching, and relocation are typically performed at a uniform spatial granularity, making zone-based decisions too coarse to support fine-grained vehicle operations. In contrast, the proposed framework ([Fig. 2b](#)) integrates an OD-based dynamic pricing scheme with a queue-jumping mechanism, which determines trip fares and queue-jumping premiums for each OD pair. Customers exhibit heterogeneous VOT and WTP, influencing their choices to either wait in the regular queue or pay a premium for priority matching. Meanwhile, vehicle dispatching and relocation decisions are formulated at the link-node level, enhancing the granularity and realism of vehicle operations.

Specifically, we consider a monopolistic ride-sourcing platform operating a fleet of fully controlled vehicles over a simulation horizon $\mathbb{T} = \{0, 1, 2, \dots, T-1\}$. The platform serves an urban region Z partitioned into several zones, and operates on a road network represented by a set of nodes (road intersections) N and a set of edges (road links) E . During each decision interval, newly generated customer requests are collected by the platform, and the demand forecasting module predicts future travel demand based on both these requests and the current system state. Let $n_{i,j,t}$ denote the total number of customers with OD pair (i, j) present in the system at time t , including both newly arriving potential customers $\hat{n}_{i,j,t}$ and those carried over from previous intervals, regardless of whether they choose to wait or jump the queue. Let $m_{i,t}$ denote the number of idle vehicles in zone i at time t . At each decision epoch, the platform determines whether to enable the queue-jumping mechanism according to the current and predicted supply-demand conditions. If enabled, customers are offered different priority levels with corresponding queue-jumping premiums. Note that the zone partition is used to characterize local supply-demand conditions, while the pricing decisions (i.e., the trip fare and queue-jumping premium) are determined at the OD level.

Customers submit their requests through the ride-sourcing app and receive the trip fare $p_{i,j,t}$ and the estimated waiting time $h_{i,j,t}$ from the platform. Upon accepting the quoted fare, they enter the matching queue and wait to be paired with an idle driver. Each customer is characterized by an individual VOT v_c and WTP w_c , representing the monetary value assigned to travel time and the maximum fare they are willing to pay, respectively. Customers with higher VOT are more likely to pay the queue-jumping premium $u_{i,j,t}$ to obtain higher matching priority and shorter expected waiting time, while those with lower VOT tend to wait until they are matched, their request times out, or they abandon the trip. Let g_c^w denote the generalized cost comprising the trip fare and waiting time

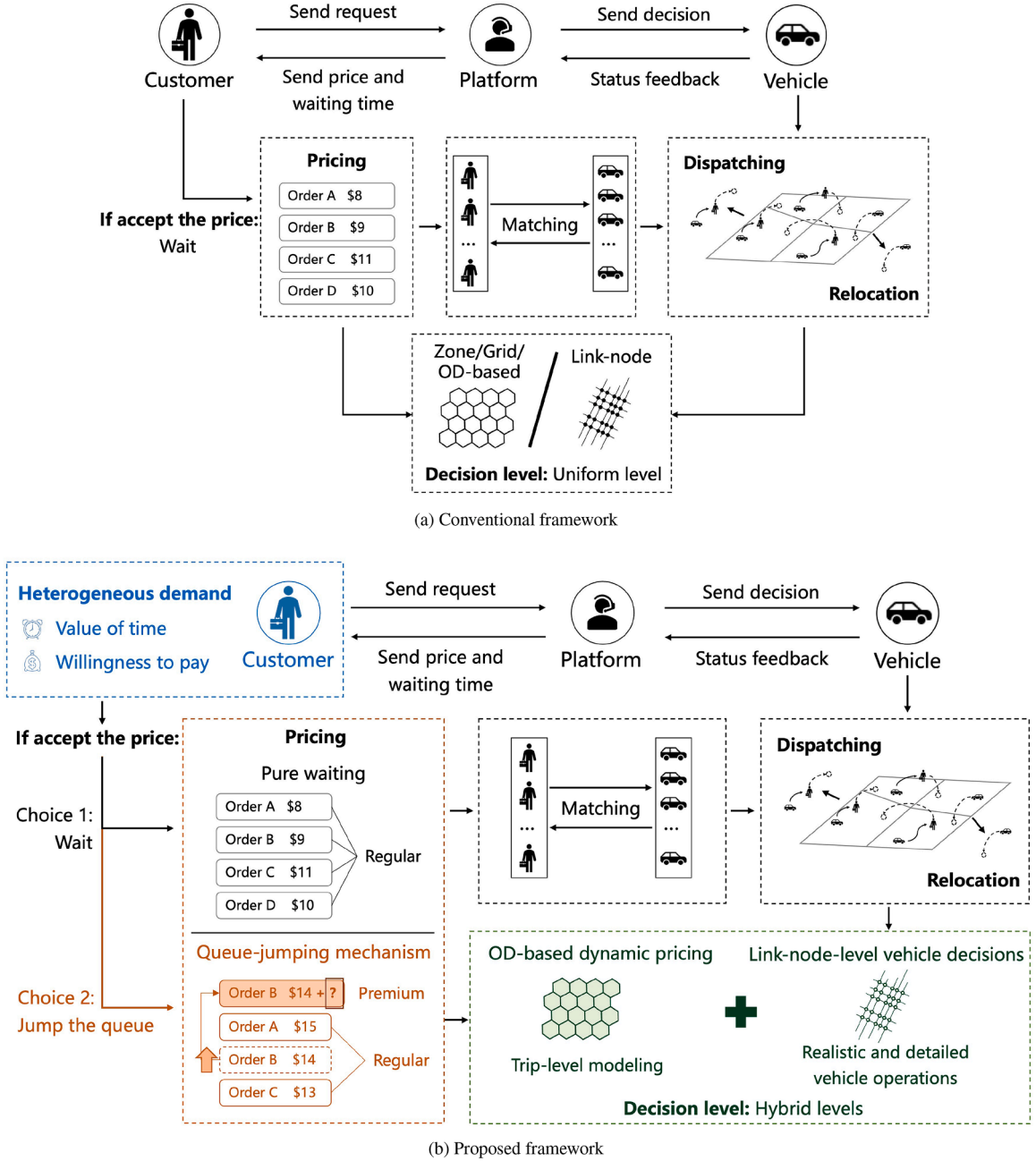


Fig. 2. Comparison of conventional and proposed ride-sourcing platform frameworks.

cost, and $g_{i,j,t}^q$ denote the total fare including the queue-jumping premium. Customers make their reservation decisions by comparing their WTP w_c with the generalized costs g_c^w and $g_{i,j,t}^q$. This decision-making process is modeled by an integral demand function, which determines the reservation demand rate $d_{i,j,t}$ under the given pricing decisions. Accordingly, among the newly arriving potential demand $\hat{n}_{i,j,t}$, only a proportion $d_{i,j,t}$ is retained by the platform, resulting in the reserved demand of $\hat{n}_{i,j,t} d_{i,j,t}$. Customers who are not matched immediately remain in the system and are carried over to subsequent decision intervals until their individual waiting tolerance is exceeded, after which they leave the system without being served.

Once customer demand has been determined, the platform dispatches vehicles to matched customers and relocates idle vehicles to better balance future supply and demand. The dispatching decision $x_{q,c,t}$ indicates whether vehicle q is assigned to customer c , and the relocation decision $y_{q,l,t}$ indicates whether vehicle q is relocated to node l .

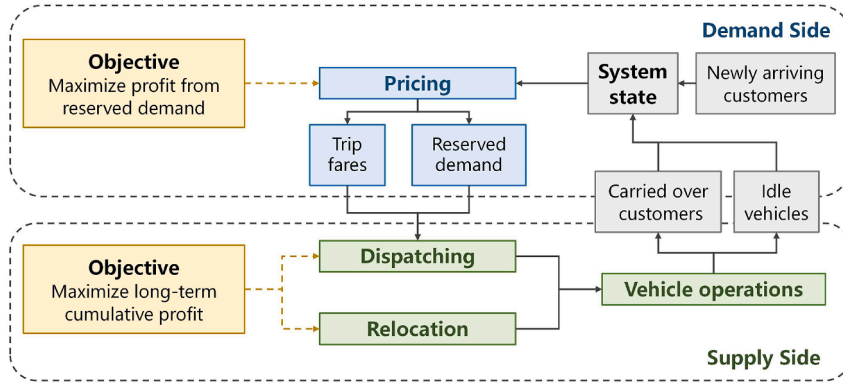


Fig. 3. Workflow of the proposed framework.

Fig. 3 illustrates the workflow of the proposed framework, which integrates dynamic pricing with vehicle dispatching and relocation in a closed loop. Customer heterogeneity in terms of VOT and WTP is incorporated through an integral demand formulation. On the demand side, the platform determines optimal pricing decisions to regulate incoming demand and maximize profits based on the current system state, represented by the number of potential customers and idle vehicles. The resulting regulated demand and corresponding trip fares are then passed as inputs to the supply side for vehicle-customer matching. On the supply side, the platform optimizes dispatching and relocation decisions to maximize long-term cumulative profit leveraging ADP. Specifically, idle vehicles are dispatched to waiting customers according to real-time supply-demand conditions, while idle vehicles are proactively relocated to anticipated high demand areas to improve system efficiency. The system state is then updated and fed back to the demand side as input for the next pricing decision, thereby forming a closed-loop interaction between demand regulation and supply optimization.

3.2. Demand side

The customer set is denoted by $\mathcal{C}$. Each customer request is represented by a tuple $c = (i, j, t_c, h_c, v_c, w_c)$, where $i, j \in \mathcal{Z}$ denote the origin and destination zones, $t_c \in \mathbb{T}$ denotes the request time, h_c denotes the maximum waiting time tolerated by the customer, v_c and w_c are the customer's VOT and WTP. Customer requests arriving during the interval $[t, t + 1)$ are aggregated and processed at time t . Each customer may either abandon the request or wait for several decision intervals until being matched or timing out.

3.2.1. Pricing mechanism

For each OD pair (i, j) at time t , the platform determines the trip fare $p_{i,j,t}$, which is calculated as the product of the price multiplier $\gamma_{i,j,t}$ and the base fare $b_{i,j,t}$. The base fare is determined according to the estimated travel time of the trip, which is consistent with the fare structure familiar to most customers.

$$p_{i,j,t} = \gamma_{i,j,t} \cdot b_{i,j,t} \quad (1a)$$

$$\gamma_{\min} \leq \gamma_{i,j,t} \leq \gamma_{\max} \quad (1b)$$

where (1b) specifies the feasible range of the price multiplier.

In addition to the monetary cost, customers also consider the time cost associated with waiting. The waiting time for each OD pair is estimated by a neural network (NN), denoted by $f_{\Omega}(\cdot)$, which captures the nonlinear supply-demand interactions:

$$h_{i,j,t} = f_{\Omega}(i, j, t, \frac{\sum_j n_{i,j,t}}{m_{i,t}}) \quad (2)$$

where $\frac{\sum_j n_{i,j,t}}{m_{i,t}}$ denotes the demand-to-supply ratio in zone i at time t , $m_{i,t}$ is the number of idle vehicles in zone i , and $n_{i,j,t}$ is the total number of customers with OD pair (i, j) in the system at time t , including both newly arriving customers and those carried over from previous decision intervals. Customers characterized by their VOT and WTP (v_c, w_c) compare the generalized costs under regular service and queue jumping to determine whether to confirm the order. The set of reserved customers is defined as

$$\Theta = \left\{ (v_c, w_c) \mid w_c \geq \min(g_c^w, g_{i,j,t}^q), 0 \leq w_c \leq \bar{w}_c, 0 \leq v_c \leq \bar{v}_c \right\} \quad (3a)$$

$$g_c^w = p_{i,j,t} + v_c \cdot h_{i,j,t} \quad (3b)$$

$$g_{i,j,t}^q = p_{i,j,t} + u_{i,j,t} \quad (3c)$$

$$u_{\min} \leq u_{i,j,t} \leq u_{\max} \quad (3d)$$

where (3b) defines the generalized cost under regular service as the sum of the trip fare and the waiting cost $v_c \cdot h_{i,j,t}$, (3c) defines the total cost under queue-jumping service as the sum of the trip fare and the queue-jumping premium $u_{i,j,t}$, and (3d) specifies the feasible

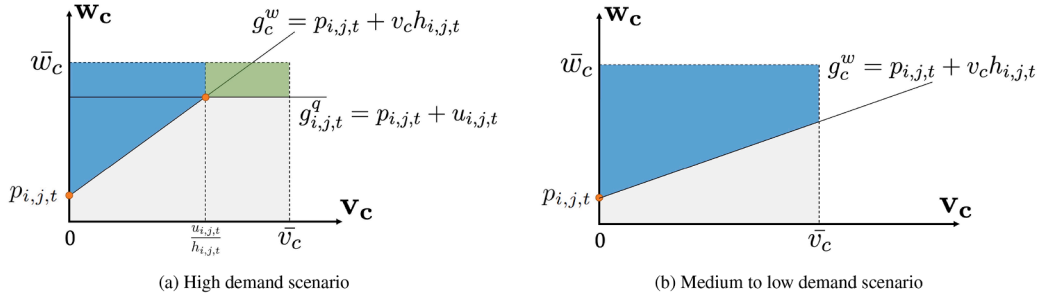


Fig. 4. Market segment in the VOT-WTP space.

range of the queue-jumping premium. The queue-jumping premium is interpreted as a payment for priority matching rather than immediate service, since premium customers may still experience a short waiting time. Instead, the premium implicitly represents the expected reduction in waiting disutility resulting from higher matching priority. Therefore, waiting time savings are not modeled as a separate cost component to avoid double counting.

Depending on the supply-demand conditions, two scenarios are considered. Note that the total customer demand considered by the platform, $n_{i,j,t}$, comprises both newly arriving demand $\hat{n}_{i,j,t}$ and carry-over demand retained from previous decision intervals. When the total customer demand originating from zone i exceeds the available vehicle supply, i.e., $\sum_j n_{i,j,t} > m_{i,t}$, the platform enables the queue-jumping option, corresponding to the high demand scenario. Otherwise, the queue-jumping option is disabled, corresponding to the medium to low demand scenario.

In high demand scenarios, the platform introduces a queue-jumping premium to differentiate customer choices. Under the optimized premium, the market is segmented into three groups, as illustrated in Fig. 4a. Customers in the blue region choose to wait, preferring to incur the time cost over the queue-jumping premium. Customers in the green region choose the queue-jumping service by paying the premium to avoid the higher waiting cost. The boundary between these two groups is defined by $g_c^w = g_c^q$, corresponding to customers with $v_c = u_{i,j,t}/h_{i,j,t}$, for whom the waiting cost equals the queue-jumping premium. Finally, customers in the gray region find both options unacceptable and abandon their travel demand. Let $d_{i,j,t}^w$ and $d_{i,j,t}^q$ denote the proportions of customers choosing regular waiting and queue jumping, respectively, with total reserved demand given by $d_{i,j,t} = d_{i,j,t}^w + d_{i,j,t}^q$. We assume that the total fare, including the queue-jumping premium, does not exceed the maximum WTP, i.e., $0 \leq u_{i,j,t} \leq \bar{w}_c - p_{i,j,t}$. This excludes pricing strategies in which no customers would choose the queue-jumping service, thereby restricting the optimization to practically meaningful solutions. It follows that

$$d_{i,j,t}^w = \int_0^{\frac{u_{i,j,t}}{h_{i,j,t}}} \int_{g_c^w}^{\bar{w}_c} D(v_c, w_c) dw_c dv_c \quad (4a)$$

$$d_{i,j,t}^q = \int_{\frac{u_{i,j,t}}{h_{i,j,t}}}^{\bar{v}_c} \int_{g_{i,j,t}^q}^{\bar{w}_c} D(v_c, w_c) dw_c dv_c \quad (4b)$$

Waiting customers pay the regular fare, whereas queue-jumping customers are charged an additional premium. The platform's objective of maximizing the profit generated from the reserved demand at time t is formulated as

$$\max_{\gamma_{i,j,t}, u_{i,j,t}} \sum_{i,j} \left[\hat{n}_{i,j,t} \cdot \left(d_{i,j,t}^w \cdot p_{i,j,t} + d_{i,j,t}^q \cdot g_{i,j,t}^q \right) \right] \quad (5)$$

$$s.t. \quad (1a), (1b), (2), (3b) - (3d), (4a), (4b)$$

In medium to low demand scenarios, i.e., $\sum_j n_{i,j,t} \leq m_{i,t}$, the platform disables the queue-jumping option and offers only regular service. Consequently, all customers pay regular fares. As illustrated in Fig. 4b, the market segmentation is simplified accordingly: the reserved demand consists solely of waiting demand $d_{i,j,t}^w$, while the queue-jumping demand $d_{i,j,t}^q = 0$. Therefore,

$$d_{i,j,t}^w = \int_0^{\bar{v}_c} \int_{g_c^w}^{\bar{w}_c} D(v_c, w_c) dw_c dv_c \quad (6a)$$

$$d_{i,j,t}^q = 0 \quad (6b)$$

Accordingly, the objective function in (5) is simplified to

$$\max_{\gamma_{i,j,t}} \sum_{i,j} \left(\hat{n}_{i,j,t} \cdot d_{i,j,t}^w \cdot p_{i,j,t} \right) \quad (7)$$

$$s.t. \quad (1a), (1b), (2), (3b), (3d), (6a), (6b)$$

3.2.2. Rolling horizon approach and event-triggered scheme

The objective functions in (5) and (7) are formulated under a myopic planning approach, considering only the profit of the current decision epoch. However, such a myopic strategy may result in suboptimal decisions. For example, during a temporary demand surge,

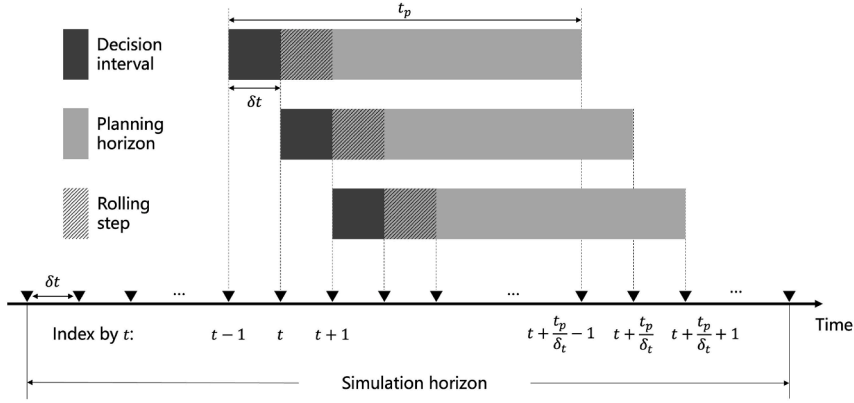


Fig. 5. Illustration of the rolling horizon approach.

the platform may impose high prices to maximize immediate revenue. If the surge is short-lived, this strategy may discourage demand reservations and reduce the platform's long-term profit. To overcome this limitation, we adopt a rolling horizon approach (see Fig. 5). The key idea is to balance responsiveness with anticipation by updating decisions at short intervals of δt (e.g., 30 s) while optimizing over a longer planning horizon t_p (e.g., 5 min). The planning horizon should be sufficiently long to capture short-term variations in supply and demand, yet short enough to limit forecasting uncertainty.

To support the rolling horizon framework, we incorporate a data-driven short-term forecasting module to predict near-future demand over the planning horizon. Rather than assuming perfect foresight, the platform infers future demand from historical observations. Specifically, customer requests are aggregated at the OD level and organized into temporal sequences, where each observation represents the total demand within a rolling time interval. These sequences are used to train a recurrent neural network (RNN)-based predictor that maps recent demand trajectories to future OD demand over the planning horizon, capturing both temporal patterns (e.g., peak and off-peak demand) and spatial correlations. At each decision epoch, the platform updates the input sequence with the most recent observations and feeds it into the trained RNN model. The resulting multi-step demand forecasts replace the unknown future demand in the optimization model and serve as inputs to the pricing, dispatching, and relocation decisions. This forecasting module enables the platform to anticipate short-term demand fluctuations and make forward-looking decisions under demand uncertainty.

Accordingly, the waiting time expression in (2) is updated as

$$h_{i,j,t} = f_{\theta} \left(i, j, t, \frac{\sum_j n_{i,j,t} + \sum_{\tau=t+1}^{t+\frac{t_p}{\delta t}} \sum_j \hat{n}_{i,j,\tau}}{\sum_{\tau=t}^{t+\frac{t_p}{\delta t}} m_{i,\tau}} \right) \quad (8)$$

where $n_{i,j,t} + \sum_{\tau=t+1}^{t+\frac{t_p}{\delta t}} \hat{n}_{i,j,\tau}$ represents the total customer demand with OD (i, j) over the planning horizon, including the current demand and the forecasted arrivals.

Under the rolling horizon framework, whether the queue-jumping option is enabled is determined using the predicted supply-demand condition over the planning horizon. Specifically, the platform enables the queue-jumping option when the predicted total demand exceeds the available vehicle supply, i.e., $\sum_j n_{i,j,t} + \sum_{\tau=t+1}^{t+\frac{t_p}{\delta t}} \sum_j \hat{n}_{i,j,\tau} > \sum_{\tau=t}^{t+\frac{t_p}{\delta t}} m_{i,\tau}$, corresponding to the high demand scenario. Otherwise, the queue-jumping option is disabled, corresponding to the medium to low demand scenario. For high demand scenarios, the rolling horizon pricing problem that maximizes the total profit from reserved demand is formulated as

$$(\gamma_{i,j,t}^*, u_{i,j,t}^*) = \arg \max_{\gamma_{i,j,t}, u_{i,j,t}} \sum_{i,j} \left[\left(n_{i,j,t} + \sum_{\tau=t+1}^{t+\frac{t_p}{\delta t}} \hat{n}_{i,j,\tau} \right) \left(d_{i,j,t}^{wv} \cdot p_{i,j,t} + d_{i,j,t}^q \cdot g_{i,j,t}^q \right) \right] \quad (9)$$

s.t. (1a), (1b), (3b) – (3d), (4a), (4b), (8)

where $\gamma_{i,j,t}^*$ and $u_{i,j,t}^*$ denote the optimal pricing decisions at time t .

For medium to low demand scenarios, the corresponding rolling horizon pricing problem is formulated as

$$\gamma_{i,j,t}^* = \arg \max_{\gamma_{i,j,t}} \sum_{i,j} \left[\left(n_{i,j,t} + \sum_{\tau=t+1}^{t+\frac{t_p}{\delta t}} \hat{n}_{i,j,\tau} \right) d_{i,j,t}^{wv} \cdot p_{i,j,t} \right] \quad (10)$$

s.t. (1a), (1b), (3b), (3d), (6a), (6b), (8)

Solving the pricing problems (9) and (10) at every decision epoch can be computationally expensive. Moreover, frequent optimization may cause prices to fluctuate substantially in response to short-term supply-demand variations, leading to market instability

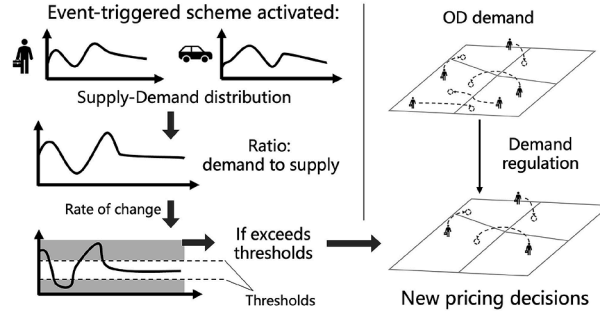


Fig. 6. Illustration of the event-triggered scheme.

that is undesirable for both drivers and customers. To address these concerns, an event-triggered scheme (see Fig. 6) is adopted, under which pricing decisions are updated only when the supply-demand condition changes significantly. An increase or decrease in the supply-demand condition is regarded as an event, and its magnitude is measured by the threshold parameter $\bar{\eta}$. Let $\eta_{i,t}$ denote the supply-demand change in zone i at time t .

$$\eta_{i,t} = \frac{\left(\sum_j n_{i,j,t} + \sum_{\tau=t+1}^{t+\frac{t_p}{\delta t}} \sum_j \hat{n}_{i,j,\tau} \right) / \left(\sum_{\tau=t}^{t+\frac{t_p}{\delta t}} m_{i,\tau} \right)}{\left(\sum_j n_{i,j,t-1} + \sum_{\tau=t}^{t+\frac{t_p}{\delta t}-1} \sum_j \hat{n}_{i,j,\tau} \right) / \left(\sum_{\tau=t-1}^{t+\frac{t_p}{\delta t}-1} m_{i,\tau} \right)} \quad (11)$$

where $\eta_{i,t}$ is defined as the demand-to-supply ratio over the planning horizon $[t, t + \frac{t_p}{\delta t}]$ divided by that over the previous planning horizon $[t-1, t + \frac{t_p}{\delta t} - 1]$.

Note that (10) is a special case of (9) with $u_{i,j,t}^* = 0$. Accordingly, the optimal pricing decisions under the event-triggered scheme are given by

$$(\bar{\gamma}_{i,j,t}^*, \bar{u}_{i,j,t}^*) = \begin{cases} (\gamma_{i,j,t-1}, u_{i,j,t-1}), & \text{if } \frac{1}{\bar{\eta}} \leq \eta_{i,t} \leq \bar{\eta} \\ (\gamma_{i,j,t}^*, u_{i,j,t}^*), & \text{otherwise} \end{cases} \quad (12)$$

(12) indicates that pricing decisions are retained from the previous decision epoch when the change in the supply-demand condition falls within the predefined threshold range. Otherwise, the pricing optimization problem is re-solved, and the updated pricing decisions are implemented.

The objective functions in (9) and (10) involve demand rates expressed as double integrals, making the optimization problems highly nonlinear and non-convex. To address this computational challenge, a discrete approximation scheme is adopted. Specifically, the continuous VOT-WTP space is partitioned into a finite number of small grids, each representing a subregion of the joint probability density. The double integrals are approximated by summing the weighted probabilities over these grids, thereby obtaining estimates of the demand rates. The price multiplier $\gamma_{i,j,t}$ and queue-jumping premium $u_{i,j,t}$ are discretized into finite levels within their respective feasible ranges. The demand rates and objective value are evaluated for each resulting pricing combination, and the combination that maximizes the profit is selected as the optimal pricing strategy.

3.3. Supply side

To provide a clearer illustration of the supply-side decisions, the customer sets are defined as follows.

$$\mathbb{C}_t = \mathbb{C}_{t-1}^w \cup \mathbb{C}_{t-1}^q \cup \mathbb{C}_t^w \cup \mathbb{C}_t^q \quad (13a)$$

$$\mathbb{C}_t^w = \left\{ c : t-1 < t_c \leq t, g_c^w \leq w_c \leq g_{i,j,t}^q, c \in \mathbb{C}_t^{new} \right\} \quad (13b)$$

$$\mathbb{C}_t^q = \left\{ c : t-1 < t_c \leq t, g_{i,j,t}^q \leq w_c < g_c^w, c \in \mathbb{C}_t^{new} \right\} \quad (13c)$$

$$\mathbb{C}_{t-1}^w = \left\{ c : t_c \leq t-1, t \leq t_c + h_c, g_c^w \leq w_c \leq g_{i,j,t}^q, c \in \mathbb{C}_{t-1} \setminus \mathbb{C}_{t-1}^m \right\} \quad (13d)$$

$$\mathbb{C}_{t-1}^q = \left\{ c : t_c \leq t-1, t \leq t_c + h_c, g_{i,j,t}^q \leq w_c < g_c^w, c \in \mathbb{C}_{t-1} \setminus \mathbb{C}_{t-1}^m \right\} \quad (13e)$$

where (13a) indicates that the total customer set $\mathbb{C}_t$ at time t consists of four subsets. $\mathbb{C}_t^w$ and $\mathbb{C}_t^q$ denote newly arriving customers who choose to wait and jump the queue, respectively, based on the generalized cost (13b)-(13c). $\mathbb{C}_t^{new} = \{c : t-1 < t_c \leq t\}$ denotes the set of newly arriving potential customers. $\mathbb{C}_{t-1}^w$ and $\mathbb{C}_{t-1}^q$ denote the waiting and queue-jumping customers from the previous decision epoch whose reservations remain active at time t , respectively (13d)-(13e). $\mathbb{C}_{t-1}$ denotes the customer set at the previous decision epoch, and $\mathbb{C}_{t-1}^m = \{c : \sum_q x_{q,c,t} = 1, c \in \mathbb{C}_{t-1}\}$ denotes the set of customers matched in the previous decision epoch.

Remark 1. Note that, in the proposed framework, customers who choose to jump the queue do not receive instantaneous service. Instead, they are assigned to a priority queue that is served ahead of the regular waiting queue but remains constrained by limited vehicle availability. When multiple customers choose the queue-jumping option within the same time period, congestion may arise within the priority queue itself, and their waiting times are determined by the relative priority ordering and prevailing supply conditions. Therefore, the queue-jumping mechanism affects the service order and relative waiting times among customers, rather than eliminating waiting altogether.

Demand-side decisions regulate customer participation and priority levels within the queueing system. The resulting customer sets, including both regular waiting and priority (queue-jumping) customers, are then passed to the supply-side module, where vehicle dispatching and relocation decisions are carried out.

3.3.1. Dispatching and relocation decisions

The vehicle set is denoted by $\mathbb{Q}$. Each vehicle is represented by a tuple $q = (o_q, h_q) \in \mathbb{Q}$, which indicates that the vehicle corresponds to its current location and time. For an occupied vehicle, o_q is the destination node and h_q is the estimated arrival time. Therefore, (o_q, h_q) represents the spatial-temporal point at which the vehicle becomes idle for future dispatching.

Leveraging historical trip records, we develop a hybrid travel-time prediction model consisting of a physics-based baseline and a neural-network correction module. Specifically, the baseline travel time is estimated using the distance-based formulation $\tau_{o,d} = \frac{L_{o,d}}{\bar{v}}$, where o and d denote the origin and destination coordinates, respectively, $L_{o,d}$ is the network distance between them, and $\bar{v}$ is a reference speed. A neural network $f_{\Delta}(\cdot)$ then learns a multiplicative correction from the input features $(t, o, d, L_{o,d})$, where t denotes the departure time. The resulting travel time estimate is

$$\hat{\tau}_{o,d,t} = \tau_{o,d} \cdot (1 + f_{\Delta}(t, o, d, L_{o,d})) \quad (14)$$

where $f_{\Delta}(\cdot)$ has a bounded output to ensure that the correction remains within a reasonable range. This hybrid formulation preserves the physical relationship between travel distance and travel time while capturing complex spatial-temporal variations, such as time-of-day effects and traffic congestion, that are difficult to model analytically.

As vehicles are assumed to be fully controlled by the platform, the decision over the supply side contains the dispatching decision $x_{q,c,t}$ and the relocation decision $y_{q,l,t}$. Dispatching decision $x_{q,c,t}$ indicates whether vehicle q is assigned to customer c at time t , which is constrained by

$$\sum_c x_{q,c,t} \leq 1 \quad (15a)$$

$$\sum_q x_{q,c,t} \leq 1 \quad (15b)$$

$$x_{q,c,t} \in \{0, 1\} \quad (15c)$$

where (15a) ensures that each vehicle is assigned to at most one customer, (15b) ensures that each customer is matched to at most one vehicle, and (15c) defines the binary dispatching decision. Specifically, $x_{q,c,t} = 1$ if vehicle q is assigned to customer c at time t , and $x_{q,c,t} = 0$ otherwise.

Relocation decision $y_{q,l,t}$ indicates whether vehicle q is relocated to node l at time t . For a vehicle $q \in \mathbb{Q}$, the decision is constrained by

$$\sum_c x_{q,c,t} + \sum_l y_{q,l,t} \leq 1 \quad (16a)$$

$$\hat{\tau}_{o_q,l,t} \leq \delta t \quad (16b)$$

$$y_{q,l,t} \in \{0, 1\} \quad (16c)$$

where (16a) ensures that a vehicle is either assigned to a customer or relocated, (16b) requires the estimated travel time from the vehicle's current idle location o_q to the relocation node l to be no greater than one decision interval δt , and (16c) defines the binary relocation decision. Specifically, $y_{q,l,t} = 1$ if vehicle q is relocated to node l at time t , and $y_{q,l,t} = 0$ otherwise.

The immediate profit associated with each decision is defined as

$$r(x_{q,c,t}) = \begin{cases} p_c - e \cdot \hat{\tau}_{o_q,o_c,t}, & c \in \mathbb{C}_t^w \cup \mathbb{C}_{t-1}^w \\ g_c^q - e \cdot \hat{\tau}_{o_q,o_c,t}, & c \in \mathbb{C}_t^q \cup \mathbb{C}_{t-1}^q \end{cases} \quad (17a)$$

$$r(y_{q,l,t}) = -e \cdot \hat{\tau}_{o_q,l,t} \quad (17b)$$

where (17a) gives the profit associated with a dispatching decision, calculated as the fare collected from customers minus the travel cost incurred from the vehicle's origin o_q to the customer's origin o_c . The fare is either the regular trip fare p_c for waiting customers or the total fare g_c^q for queue-jumping customers. Here, e denotes the travel cost per unit time. (17b) gives the profit associated with a relocation decision, defined as the negative travel cost incurred when a vehicle relocates from its origin o_q to node l .

The platform's objective is to maximize the expected total profit over the planning horizon

$$\max_{x_{q,c,t}, y_{q,l,t}} \mathbb{E} \left\{ \sum_t \left(\sum_q \sum_c r(x_{q,c,t}) + \sum_q \sum_l r(y_{q,l,t}) \right) \right\} \quad (18)$$

While dynamic programming is a standard approach for solving (18), it is challenging to apply it directly due to the curse of dimensionality arising from the high-dimensional state and decision spaces. Therefore, we adopt an approximate dynamic programming (ADP) framework (Powell, 2007; Liang et al., 2021), in which the value function in the Bellman equation is approximated by a parameterized function learned from data. This enables efficient policy learning for dispatching and relocation decisions while maintaining computational tractability. We next formulate the ADP model by defining the state, decision variables, state transition, and objective function.

State The state at time t is represented by the vehicle's spatial-temporal information

$$s_t = (o_q, h_q) \quad (19)$$

Decision The decision at time t consists of dispatching and relocation actions

$$a_t = (x_{q,c,t}, y_{q,l,t}) \quad (20)$$

where the dispatching assigns idle vehicles to customers to efficiently match supply and demand while minimizing service delays, whereas the relocation action moves idle vehicles to better satisfy future spatial-temporal demand.

State transition Given the dispatching and relocation decisions, the system state evolves according to

$$s_{t+1} = M(s_t, a_t) \quad (21)$$

In principle, the state transition is governed by the conditional probability distribution $\mathbb{P}(s_{t+1} | s_t, a_t)$, which captures uncertainties in travel times, demand realizations, and matching outcomes. However, explicitly modeling or estimating this transition distribution is computationally intractable in large-scale network settings due to the high dimensionality and complex dependencies involved. Instead, we adopt a data-driven transition model in which $M(\cdot)$ is implicitly learned from historical trip data and simulation. This enables the proposed ADP framework to approximate the underlying stochastic dynamics without explicitly specifying the transition probabilities, thereby improving computational tractability while preserving modeling flexibility.

Objective function Given the current state s_t and decision a_t , the platform receives an immediate reward $r(s_t, a_t)$, which represents the profit obtained from dispatching and relocation decisions. Under a policy π , the objective is to maximize the long-term cumulative profit in the planning horizon

$$V^\pi(s_0) = \mathbb{E}_\pi \left[\sum_{t=0}^T r(s_t, a_t) \mid s_0 \right] \quad (22)$$

For an arbitrary state s_t , the corresponding value function is defined as

$$V^\pi(s_t) = \mathbb{E}_\pi \left[\sum_{k=t}^T r(s_k, a_k) \mid s_t \right] \quad (23)$$

According to the Bellman optimality equation, the optimal value function satisfies

$$V^*(s_t) = \max_{a_t} \mathbb{E} [r(s_t, a_t) + \alpha V^*(s_{t+1}) \mid s_t, a_t] \quad (24)$$

where α is the discount factor. Equivalently,

$$Q(s_t, a_t) = \mathbb{E} [r(s_t, a_t) + \alpha V^*(s_{t+1}) \mid s_t, a_t] \quad (25)$$

and the optimal policy is obtained by

$$\pi^*(s_t) = \arg \max_{a_t} Q(s_t, a_t) \quad (26)$$

For the dispatching decision, the objective function can be expressed as

$$\begin{aligned} \max_{x_{q,c,t}} \sum_{q \in \mathbb{Q}} \sum_{c \in \mathbb{C}_t} Q(s_t, a_t) \cdot x_{q,c,t} \\ \text{s.t.} \quad (15a) - (15c) \end{aligned} \quad (27)$$

For the relocation decision, the objective function can be expressed as

$$\begin{aligned} \max_{y_{q,l,t}} \sum_{q \in \mathbb{Q}} \sum_{l \in \mathbb{N}} Q(s_t, a_t) \cdot y_{q,l,t} \\ \text{s.t.} \quad (16a) - (16c) \end{aligned} \quad (28)$$

To mitigate the curse of dimensionality in dynamic programming, we employ value function approximation. Specifically, a neural network is used to approximate the value function $V(s_t)$ by $\hat{V}(s_t | \theta)$, where θ denotes the neural network parameters. The network is trained using temporal-difference (TD) learning:

$$\delta_t^{(i)} = r(s_t, a_t) + \alpha \cdot \hat{V}(s_{t+1} | \theta) - \hat{V}(s_t | \theta) \quad (29a)$$

$$L(\theta) = \frac{1}{2} \left(\delta_t^{(i)} \right)^2 \quad (29b)$$

$$\theta_{k+1} = \theta_k - \beta \cdot \nabla_\theta \cdot L(\theta_k) \quad (29c)$$

Algorithm 1 Dynamic pricing framework integrating vehicle dispatching and relocation decisions.**Input:** Zone set Z , customer set C , vehicle set Q

```

1: for each decision epoch  $t \in \mathbb{T}$  do
2:   Environment observe and state update:
3:   Observe the current vehicle set  $Q_t \in Q$  and newly arriving customer set  $C_t^{new}$ , and update  $m_{i,t}, n_{i,j,t}$ 
4:   Pricing decision:
5:   Predict the newly arriving customer demand  $\hat{n}_{i,j,t}$  over the planning horizon
6:   Calculate  $\eta_{i,t}$  according to (11)
7:   if  $\eta_{i,t}$  exceeds the predefined event-triggered threshold range then
8:     Estimate the waiting time  $h_{i,j,t}$  according to (8)
9:     if  $\sum_j n_{i,j,t} + \sum_{\tau=t+1}^{t+\frac{t_p}{\delta t}} \sum_j \hat{n}_{i,j,\tau} > \sum_{\tau=t}^{t+\frac{t_p}{\delta t}} m_{i,\tau}$  then ▷ High demand scenario
10:      Solve for the optimal pricing decision  $\gamma_{i,j,t}^*, u_{i,j,t}^*$  according to (9)
11:    else ▷ Medium/low demand scenario
12:      Solve for the optimal pricing decision  $\gamma_{i,j,t}^*$  according to (10)
13:    end if
14:  else
15:    Use the pricing decision  $\gamma_{i,j,t-1}, u_{i,j,t-1}$  from the previous decision epoch
16:  end if
17:  Demand update:
18:  Update the current reserved customer set  $C_t$  according to (13a)
19:  Dispatching decision:
20:  Solve for the optimal dispatching decision  $x_{q,c,t}^*$  according to (27)
21:  Relocation decision:
22:  Solve for the optimal relocation decision  $y_{q,l,t}^*$  according to (28)
23:  Environment update:
24:  Remove matched customers from  $C_t$  and update the vehicle status in  $Q_t$ 
25: end for

```

where (29a) defines the TD error, (29b) is the corresponding loss function, and (29c) updates the neural network parameters using gradient descent. Here, β denotes the learning rate.

Algorithm 1 summarizes the proposed dynamic pricing framework integrated with vehicle dispatching and relocation. At each decision epoch, the platform first observes the current system state, including vehicle states and newly arriving customer requests, and updates zone-level supply-demand indicators. It then predicts customer demand over the planning horizon. Based on the predicted demand, an event-triggered mechanism determines whether the pricing policy should be updated. When the trigger condition is satisfied, the platform estimates expected waiting time and solves the corresponding pricing problem under either high demand or medium to low demand scenarios. Otherwise, the pricing decisions from the previous decision epoch are retained to avoid excessive price fluctuations.

Based on the pricing decisions, customer choices are updated accordingly, yielding the regulated customer set that serves as the input to the subsequent operational layer. The platform then determines the dispatching and relocation decisions by solving the corresponding optimization problems to maximize the expected long-term system profit while satisfying the operational constraints. Finally, the system state is updated by removing matched customers and updating vehicle states, and the process repeats in a rolling horizon manner until the end of the operating period.

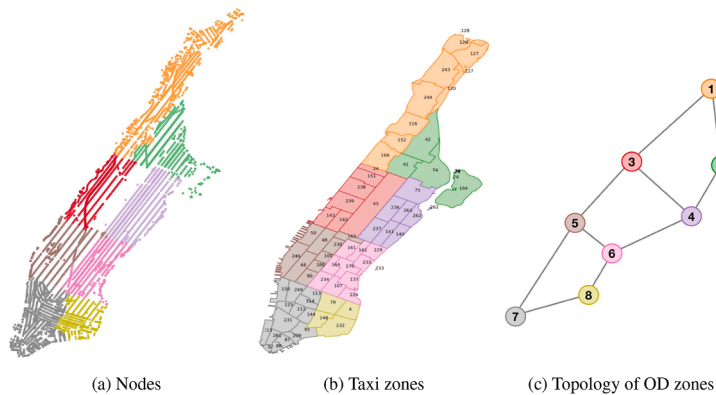


Fig. 7. Nodes and aggregated taxi OD zones in Manhattan. Adjacent taxi zones are merged into a single zone and represented by the same color.

4. Empirical study

4.1. Simulation environment

The proposed framework is evaluated using a simulator built on the Manhattan road network, with historical taxi trip records from June 2016 provided by the New York City Taxi and Limousine Commission ¹. The dataset contains pickup and drop-off coordinates, travel distance, travel time, and trip fare for each trip. The study area is partitioned into 8 zones by aggregating the original 263 taxi zones, as illustrated in Fig. 7. Pickup and drop-off locations are mapped to their nearest nodes in the underlying road network, which consists of 4486 nodes and 9755 links. A fleet of 1000 vehicles is initially randomly distributed over the network. Each simulation run starts at 00:00, while performance metrics are collected from 06:00 to 24:00 to mitigate cold-start effects. Approximately 80,000 orders are observed during the evaluation period. Historical demand data from the first 25 days are used to train the prediction and decision-making modules of the proposed framework, whereas the remaining days are reserved for out-of-sample evaluation.

Table 2 summarizes the neural network configurations used throughout the numerical experiments. Waiting time and travel time are estimated using lightweight feedforward neural networks to enable efficient training and inference. OD demand forecasting employs an RNN-based model to capture temporal correlations in demand patterns. For the ADP-based dispatching and relocation algorithm, the value function (with discount factor $\alpha = 0.9$) is approximated by a fully connected neural network with three hidden layers. Training samples are generated from simulated transitions $s_t, a_t, r(s_t, a_t)$. All models are trained using the Adam optimizer with a learning rate of 0.01 and implemented in Python with PyTorch on an Apple M1-based MacBook.

Table 2
Neural network architectures and training configurations.

Module	Architecture	Activation	Epochs	Batch Size
Waiting Time Estimation	NN (32-16) + zone embedding	ReLU	400	1000
Demand Forecasting	RNN (64-32)	Tanh	200	1000
Travel Time Estimation	NN (32-16)	ReLU	400	1000
ADP Value Approximation	NN (32-16-4)	Leaky ReLU	1000	1000

Note: Numbers in parentheses indicate the numbers of neurons in the hidden layers.

Note that the proposed integral demand formulation is not restricted to any specific functional form of the joint VOT-WTP distribution. The framework is sufficiently general to accommodate any joint distribution, provided that the corresponding probability density or cumulative distribution functions are well defined. In line with Zhang et al. (2018), a truncated bivariate normal distribution is adopted to characterize customer heterogeneity in VOT and WTP:

$$D(v_c, w_c) \sim \mathcal{N}(\mu, \Sigma), \quad \mu = (24, 18)^\top, \quad \Sigma = \begin{pmatrix} 64 & 0 \\ 0 & 36 \end{pmatrix} \quad (30)$$

where the mean VOT is set to $\mu(v_c) = 24$ USD/hour, which is consistent with the estimated average VOT of 24.47 USD/hour in the United States reported by Gao et al. (2019). The corresponding standard deviation is $\sigma(v_c) = 8$ USD/hour. The mean WTP is set to $\mu(w_c) = 18$ USD with a standard deviation of $\sigma(w_c) = 6$ USD.

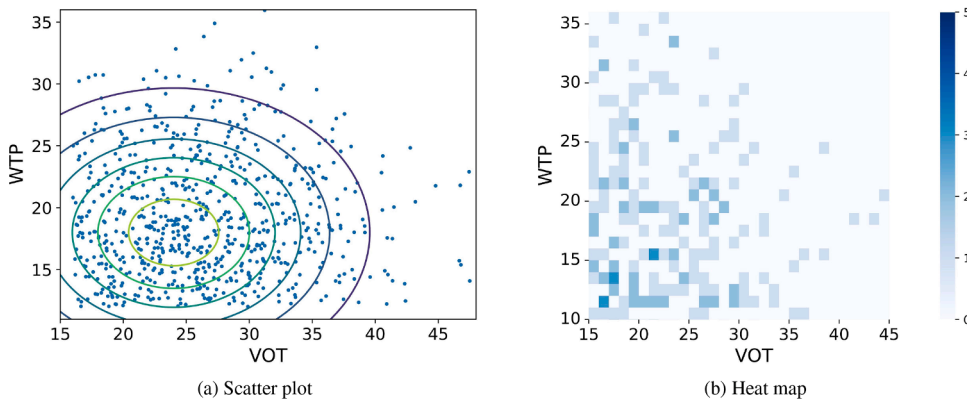


Fig. 8. Heterogeneous demand modeled by a truncated bivariate normal distribution.

In general, individual incomes exhibit a long-tailed distribution, with most individuals concentrated at lower income levels and a relatively small proportion of high-income earners raising the overall mean (Neal and Rosen, 2000). The distribution of VOT exhibits a

¹ NYC TLC opendata: <https://www1.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

similar characteristic. Customers with different VOT and WTP levels naturally exhibit heterogeneous travel preferences. Accordingly, we focus on customers whose VOT exceeds the median, as they represent a substantial proportion of the ride-sourcing market and are more likely to be time-sensitive and willing to pay a premium for improved service. Based on this consideration, the bivariate normal distribution of VOT and WTP is truncated to $v_c \in [16, 48]$ USD/hour and $w_c \in [12, 36]$ USD, respectively, as shown in Fig. 8a. Each sampled point represents an individual customer characterized by a unique combination of VOT and WTP values. The corresponding density is illustrated in Fig. 8b, where darker blue shades indicate regions with higher customer density.

4.2. Simulation results analysis

In this section, we comprehensively evaluate the performance of the proposed framework from multiple perspectives. Specifically, we examine its performance under different demand patterns in Section 4.2.1, assess the impact of integrating the queue-jumping option into the dynamic pricing framework in Section 4.2.2, evaluate the benefits of the rolling horizon approach in Section 4.2.3, and investigate the effectiveness of the event-triggered scheme in Section 4.2.4. Finally, we compare the proposed ADP solution with other alternative approaches in Section 4.2.5.

4.2.1. Performance under different demand patterns

To investigate the effects of different demand patterns, we consider two additional cases: homogeneous demand (HD) and finite discrete demand (FDD). Given the mean values of VOT and WTP, HD is modeled as a single representative customer type with (24, 18). FDD is approximated by three discrete customer groups with VOT and WTP levels below, equal to, and above the mean, represented by (18, 12), (24, 18), and (30, 24), respectively. The heterogeneous demand is modeled by a truncated bivariate normal distribution (TBND) defined in (30).

Table 3
Performance under different demand patterns.

Demand type		HD	FDD	TBND
Profit (USD)		413,236	513,858	648,462
Trip fare		486,003	591,865	727,732
Queue-jumping premium		1,406	4,521	15,673
Routing consumption		74,173	82,528	94,943
Abandonment		37,644	35,013	17,130
Realized orders		42,356	44,987	62,677
Reserved orders		42,356	44,987	62,870
Waiting ^a	Realized	41,465	42,773	44,697
	Reserved	41,465	42,773	44,890
Queue-jumping ^b	Realized	891	2,214	17,980
	Reserved	891	2,214	17,980

^a Number of waiting customers.

^b Number of queue-jumping customers.

^c Note: The same interpretation applies to all subsequent tables.

Table 3 summarizes the performance of the proposed framework under different demand pattern assumptions. Compared with the HD scenario, the FDD setting yields moderate improvements in both reserved and realized orders. Meanwhile, the number of customers choosing the queue-jumping option increases by nearly 150%, leading to a 24.3% increase in system profit. This improvement

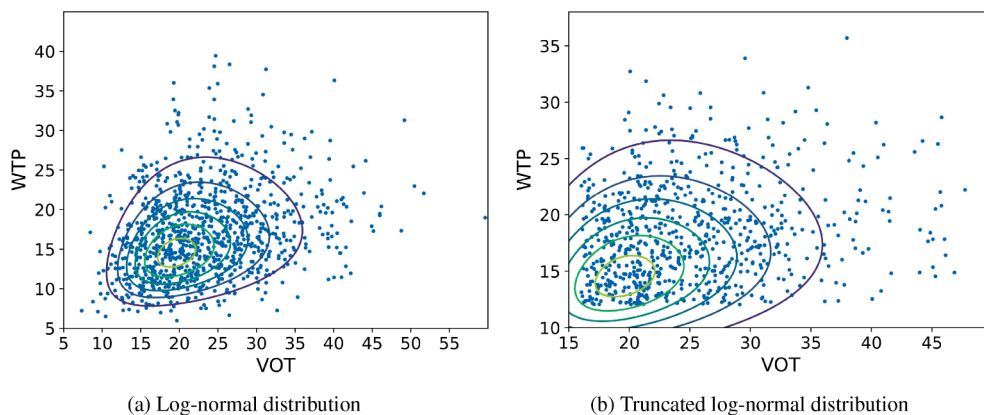


Fig. 9. Different log-normal demand distributions.

demonstrates the benefit of introducing representative customer types, which enables differentiated pricing and queue-jumping decisions across customer segments. Building on this, the heterogeneous demand setting, modeled by a continuous distribution of VOT and WTP, achieves the highest system profit. Significant increases in both realized and reserved orders, together with a substantial reduction in customer abandonments, contribute to higher revenues from trip fares and queue-jumping premiums. Moreover, a larger proportion of customers choose either to wait or jump the queue, indicating more effective market segmentation. Consequently, the proposed framework is able to allocate vehicle capacity more efficiently while better aligning pricing decisions with heterogeneous customer preferences. Overall, these findings highlight the importance of accurately capturing customer heterogeneity for improving operational efficiency, market segmentation, and system profitability.

To demonstrate that the proposed framework can accommodate alternative demand distributions, a bivariate log-normal distribution is adopted to capture the right-skewed and long-tailed characteristics of customer VOT and WTP, as illustrated in Fig. 9a. This distribution reflects the presence of a small proportion of customers with exceptionally high VOT and WTP values and is specified as

$$(\ln v, \ln w) \sim \mathcal{N}(\mu, \Sigma), \quad \mu = (3.1, 2.8)^\top, \quad \Sigma = \begin{pmatrix} 0.1 & 0.03 \\ 0.03 & 0.1 \end{pmatrix}$$

where v and w denote the VOT and WTP, respectively. This specification induces right-skewed marginal distributions while preserving the positive correlation between the two variables. The distribution is further truncated to $v_c \in [16, 48]$ USD/hour and $w_c \in [12, 36]$ USD, as shown in Fig. 9b.

Table 4
Performance under different log-normal demand distributions.

		LND	TLND
Profit (USD)		529,022	629,544
Trip fare		607,342	709,133
Queue-jumping premium		14,634	20,806
Routing consumption		92,954	100,395
Abandonment		23,989	15,367
Realized orders		55,782	63,211
Reserved orders		56,011	64,633
Waiting	Realized	40,534	42,727
	Reserved	40,762	44,032
Queue-jumping	Realized	15,248	20,484
	Reserved	15,249	20,601

Table 4 summarizes the performance of the proposed framework under the log-normal distribution (LND) and truncated log-normal distribution (TLND) settings. Compared with LND, TLND achieves a clear improvement in system performance. System profit increases by nearly 19%, accompanied by substantial increases in both realized and reserved orders. This improvement is primarily driven by a shift in demand composition. The truncated distribution captures a larger proportion of customers with higher VOT and WTP (see Fig. 10), which better reflects the target customer base of ride-sourcing platforms. Under TLND, a larger proportion of customers choose either to wait or to jump the queue. In particular, the number of customers selecting the queue-jumping option increases substantially, indicating a stronger willingness to pay for priority service.

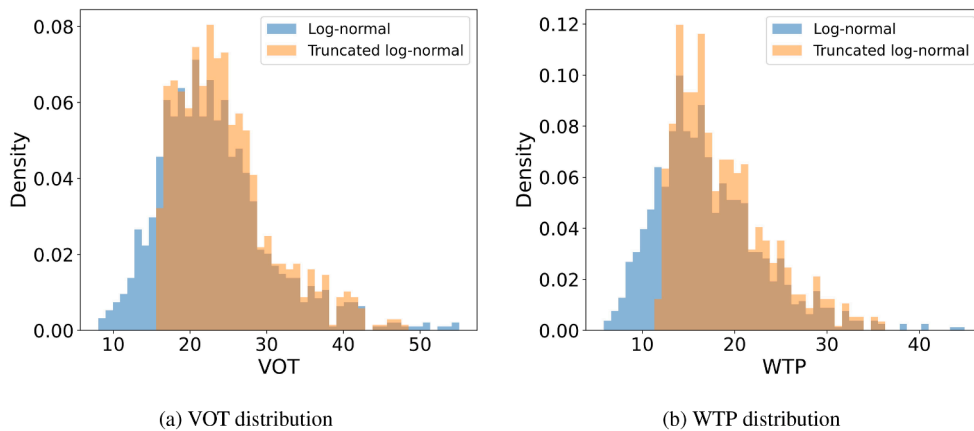


Fig. 10. Histograms of different log-normal distributions.

Overall, a truncated heterogeneous demand distribution better captures the demand characteristics of ride-sourcing platforms. By accounting for the joint distribution of VOT and WTP, the proposed framework increases the number of reserved orders, improves service for high-value customers, and ultimately enhances system profitability.

Table 5
Performance with and without queue-jumping.

		With queue-jumping	Without queue-jumping
Profit (USD)		648,462	618,801
Trip fare		727,732	715,487
Queue-jumping premium		15,673	–
Routing consumption		94,943	96,686
Abandonment		17,130	18,442
Realized orders		62,677	61,382
Reserved orders		62,870	61,558
Waiting	Realized	44,697	61,382
	Reserved	44,890	61,558
Queue-jumping	Realized	17,980	–
	Reserved	17,980	–

4.2.2. Effect of the queue-jumping option

Table 5 summarizes the performance of the proposed framework with and without the queue-jumping option. Overall, introducing the queue-jumping option leads to a modest improvement in system profit, primarily driven by an increase in fulfilled orders and additional revenue from priority services. By allowing customers with higher VOT and WTP to receive higher service priority, the queue-jumping mechanism improves the allocation efficiency of limited vehicle supply under high demand conditions, thereby partially alleviating the supply-demand imbalance.

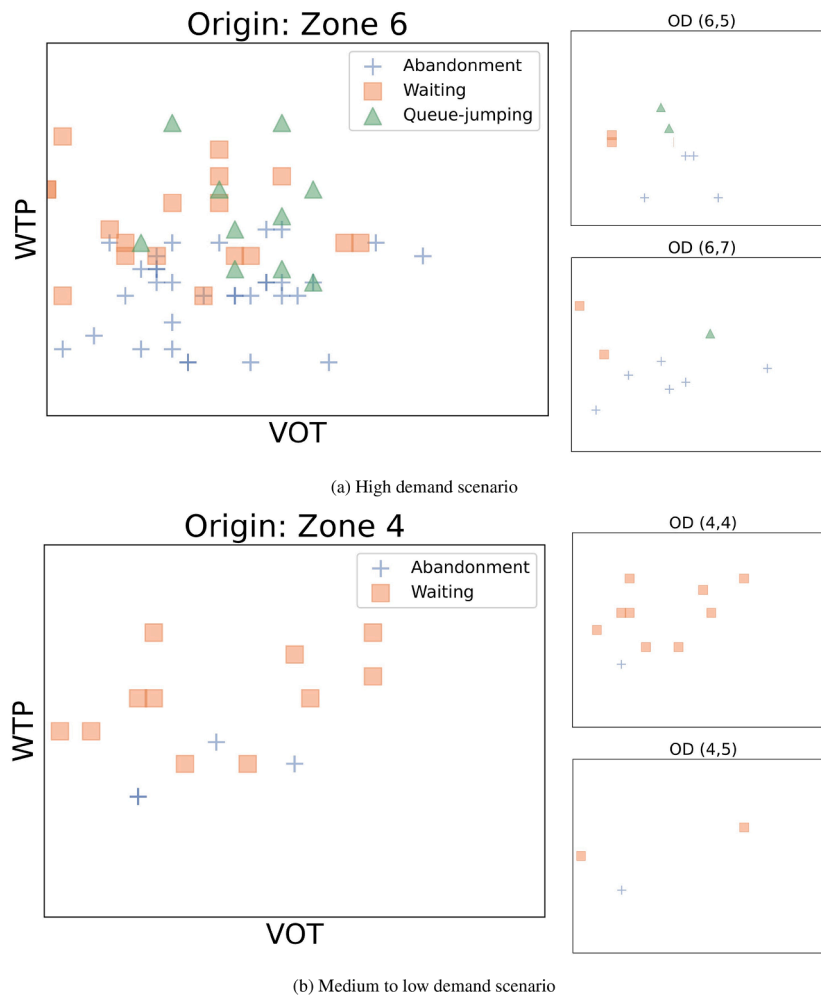


Fig. 11. Customer choices for different OD pairs within a decision interval.

Queue-jumping also redistributes service opportunities across customer groups. Fig. 11a and b illustrate customer decision-making behaviors under high demand and medium to low demand scenarios, respectively.

In high demand scenarios, a substantial proportion of customers, particularly those with medium and low VOT, choose to abandon their requests in response to elevated prices. Among those who continue to place orders, customers with low VOT and high WTP (upper-left corner in Fig. 11a) are more likely to wait than to jump the queue, as the cost of waiting is relatively low compared with the queue-jumping premium. In contrast, customers with high VOT and high WTP (upper-right corner in Fig. 11a) exhibit a stronger preference for time savings and are therefore more likely to select the queue-jumping option. These patterns are consistent with the market segmentation shown in Fig. 4a, highlighting heterogeneous behavioral responses across customer segments.

In medium to low demand scenarios, where supply-demand conditions are less constrained, some customers remain unserved. Customers with high VOT and high WTP (upper-right corner in Fig. 11b) face a trade-off between waiting and abandoning their requests. Those with relatively lower WTP are more likely to abandon, whereas those with higher WTP are more willing to wait. This behavior is consistent with practical ride-sourcing operations, where customers in areas with limited or unevenly distributed vehicle supply often decide whether to wait for service or abandon their requests.

4.2.3. Effects of rolling horizon approach

Fig. 12 illustrates the number of trip orders across different VOT levels for each zone, including the total generated demand (Origin) and the realized demand under different planning settings. Overall, demand decreases monotonically as VOT increases across all zones. Although supply constraints prevent all demand from being served, the extended planning horizon consistently realizes a larger proportion of demand than the short horizon setting. By anticipating future demand and supply conditions, it enables more strategic pricing, dispatching, and relocation decisions. Consequently, the system serves a larger proportion of demand, particularly customers with medium to low VOT, who constitute the majority of the market. This advantage is especially evident in high-demand zones (Zones 4–6), where the queue-jumping option is more frequently utilized. The extended planning horizon allows the system to better balance serving current demand with preserving capacity for future requests.

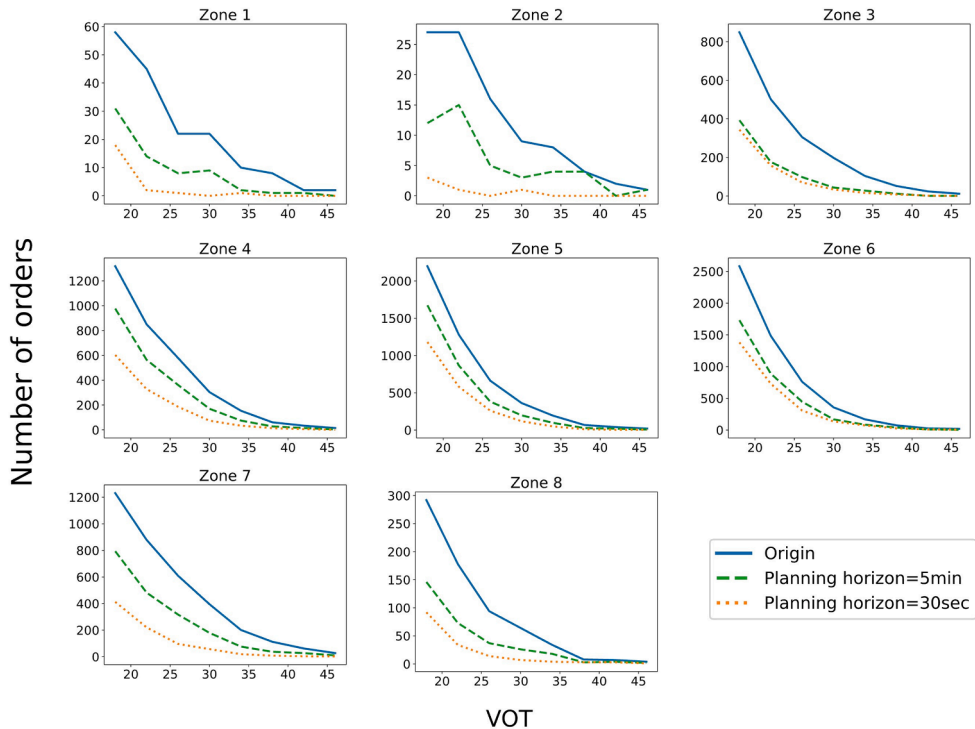


Fig. 12. Reservation and realization of demand across different planning horizons and zones.

4.2.4. Effects of event-triggered scheme

Fig. 13 presents the system profit, reservation rate, and service rate under different event-triggered thresholds. The reservation rate is defined as the ratio of reserved to total orders, while the service rate is the ratio of fulfilled to total orders. A threshold of 1.0 corresponds to the baseline without the event-triggered scheme, where pricing decisions are updated at every decision epoch. Thresholds from 1.1 to 1.3 represent increasing levels of event-triggering, resulting in less frequent pricing updates.

The event-triggered scheme generally improves the platform's service performance. However, its impact on system profit depends on the threshold. Among all tested values, a threshold of 1.1 achieves the highest system profit, reservation rate, and service rate.

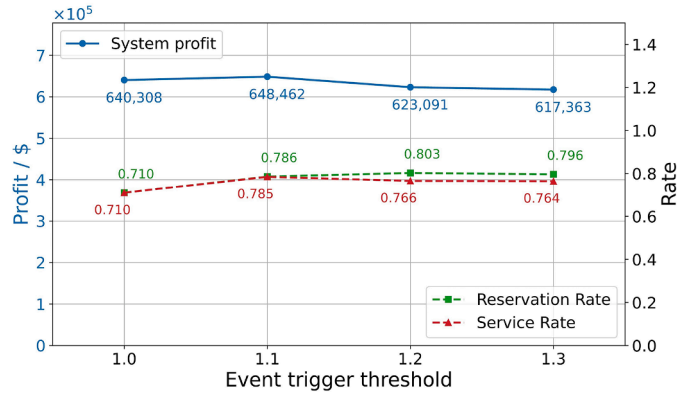
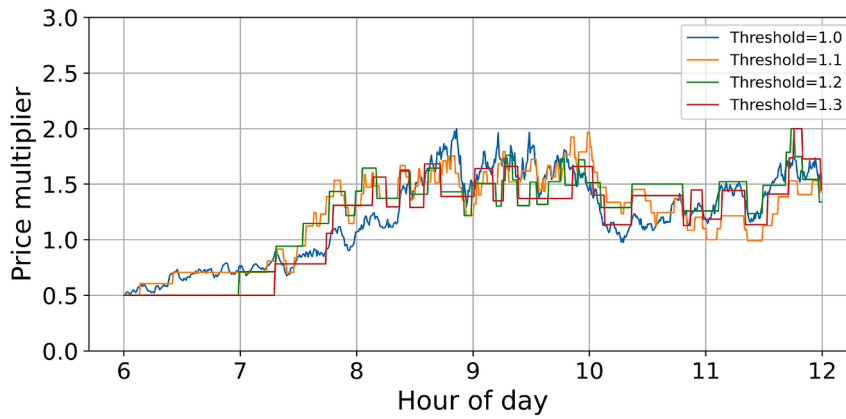
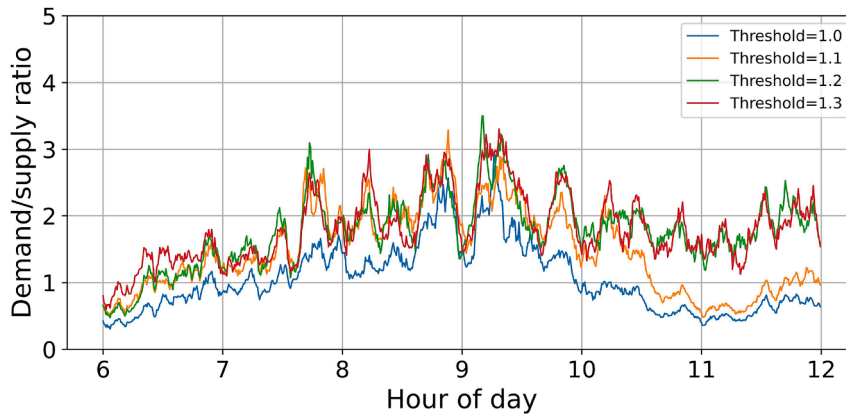


Fig. 13. Performance under different event-triggered thresholds.



(a) Price multiplier



(b) Supply-demand ratio

Fig. 14. Comparison of price multiplier and supply-demand condition under different event-triggered thresholds (OD (5,7)).

When the threshold exceeds 1.1, the reservation rate continues to increase, whereas the service rate declines. This result indicates that the platform reserves more orders than it can ultimately fulfill. The main reason is that less frequent pricing updates reduce the platform's responsiveness to supply-demand changes, as shown in Fig. 14. For example, under a threshold of 1.3, the price multiplier remains low for an extended period (see Fig. 14a). Meanwhile, demand begins to surge around 9:40 (see Fig. 14b). The delayed pricing adjustment leads to suboptimal pricing decisions and lower system profit.

These results suggest that a threshold of 1.1 provides a good balance between stabilizing price fluctuations and maintaining responsiveness to dynamic supply-demand conditions. Consequently, it achieves the best overall performance on the Manhattan network. The optimal threshold may vary across networks and requires further analysis for other settings.

4.2.5. Comparison between ADP and alternative approaches

To evaluate the effectiveness of the proposed dispatching and relocation module, the ADP approach is compared with two reinforcement learning approaches: MARL-TD3 and GraphDQN.

- **MARL-TD3:** A multi-agent reinforcement learning framework built upon Twin Delayed Deep Deterministic Policy Gradient (TD3) (Fujimoto et al., 2018). Vehicles are modeled as cooperative agents that generate continuous dispatching and relocation actions in the coordinate space. The policy is trained to maximize the cumulative system reward, thereby improving long-term operational performance.
- **GraphDQN:** A graph-based deep Q-network in line with Yu and Hu (2022), where the transportation network is represented as a graph and node embeddings are learned through graph convolution layers. Based on the aggregated spatial features, the Q-network determines dispatching and relocation actions while explicitly capturing inter-zone spatial dependencies.

For a fair comparison, all methods share the same simulation environment, demand realization process, state representation, and reward function. The hyperparameters of the baseline methods are carefully tuned to ensure stable training and convergence.

Table 6
Performance comparison with baseline methods.

Method	Profit (\$)	Service Rate (%)	Avg. Waiting Time (s)	Avg. Empty VMT (mile)	Computation Time (min)
MARL-TD3	652,338	80.9	188.6	86.1	28.6
GraphDQN	631,840	77.1	181.3	81.3	36.9
ADP (Ours)	648,462	78.5	167.7	77.8	24.3

Table 6 summarizes the performance of MARL-TD3, GraphDQN, and the proposed ADP approach. Among the three methods, MARL-TD3 achieves the highest profit and service rate, demonstrating the effectiveness of multi-agent coordination in improving long-term operational performance. However, these gains come at the expense of longer customer waiting times, higher empty vehicle miles traveled (VMT), and substantially greater computational cost. GraphDQN incorporates the spatial structure of the road network through graph convolution, but the additional model complexity leads to considerable computational overhead while providing only marginal improvements in operational performance.

In comparison, the proposed ADP approach achieves profit and service rate comparable to those of MARL-TD3 while substantially reducing customer waiting times, empty VMT, and computational cost. This balanced performance indicates that the proposed method effectively coordinates dispatching and relocation decisions while maintaining high computational efficiency. The lower waiting times and empty VMT also suggest more efficient vehicle utilization and improved matching between supply and demand. Overall, these results demonstrate that the proposed approach provides a favorable trade-off between service quality, operational efficiency, and computational complexity, making it well suited for large-scale ride-sourcing applications.

4.3. Discussion on model robustness

4.3.1. Effects of spatial partitioning

The granularity of spatial partitioning influences both the computational efficiency of the framework and the profitability of the platform. Since dispatching determines vehicle-customer matching and routing decisions, it is performed at the link-node level, where spatial differences can substantially affect pickup time, detour distance, and matching outcomes.

In contrast, the pricing module is designed to regulate aggregate supply-demand imbalance and customer competition within a zone rather than optimize individual matches. Consequently, pricing does not require the same level of spatial resolution. Applying pricing at the link-node level would result in overly sparse demand within each unit, reducing meaningful customer competition and limiting the effectiveness of dynamic pricing. In addition, the resulting waiting time and demand estimates would become increasingly noisy, leading to higher computational complexity without corresponding performance gains.

Aggregating links and nodes into larger pricing zones increases demand density and preserves meaningful competition among customers. However, using overly coarse partitions, such as treating the entire city as a single zone, fails to capture spatial heterogeneity and local supply-demand imbalances. Therefore, the number of pricing zones represents a non-monotonic trade-off between spatial resolution and estimation reliability. Increasing the number of zones initially improves system performance by capturing localized market conditions more accurately, whereas excessive partitioning eventually degrades performance because of demand sparsity and unreliable estimates. To quantify this trade-off, a one-at-a-time sensitivity analysis is conducted.

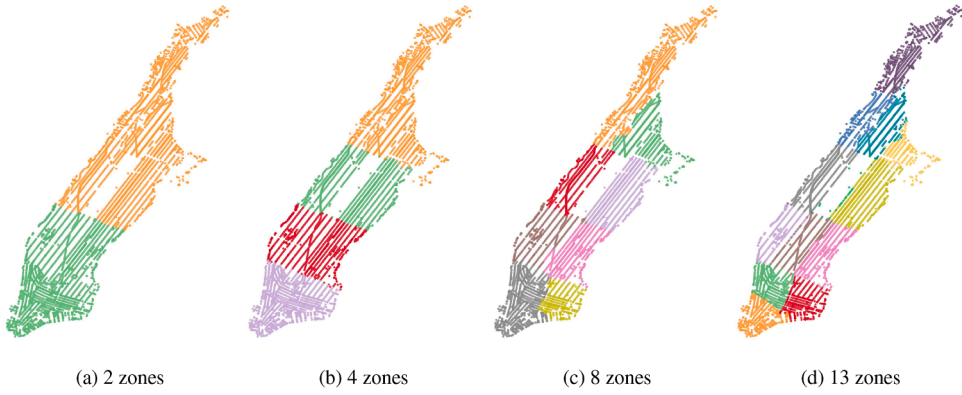


Fig. 15. Topology of different OD zones in Manhattan.

Fig. 16 presents the system performance under different spatial partitions, with the corresponding zone layouts shown in Fig. 15. Increasing the number of pricing zones from 2 to 8 leads to noticeable improvements in both profit and service rate, suggesting that the pricing strategy under a finer spatial partition better captures localized supply-demand conditions. However, further increasing the number of zones to 13 degrades system performance, with both profit and service rate declining. As shown in Fig. 16c, the computational time increases steadily as the number of zones grows. These results indicate that, for the Manhattan network considered in this study, an 8-zone partition achieves a favorable balance between operational performance and computational efficiency.

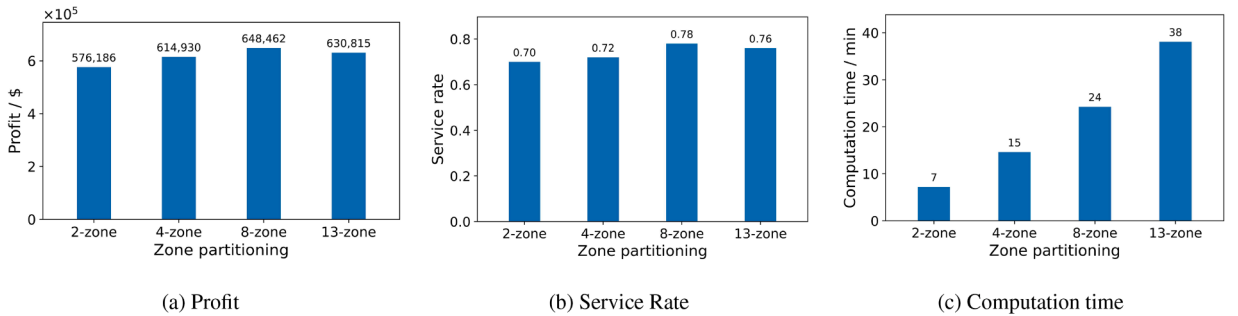


Fig. 16. Performance under different spatial partitioning.

4.3.2. Driver behavior

The current framework assumes full driver compliance with platform dispatching and relocation decisions, a common assumption in centralized ride-sourcing optimization. In practice, ride-sourcing platforms employ various incentive and penalty mechanisms to promote driver compliance. Acceptance-rate requirements and temporary account suspensions are commonly used, particularly for dispatched orders. Nevertheless, drivers may still reject assigned requests in real-world operations, introducing additional uncertainty into system performance.

To evaluate the robustness of the proposed framework under driver non-compliance, scenarios with driver rejection rates of 0%, 5%, 10%, or 15% are considered. Rejected requests are returned to the matching pool, potentially causing additional delays or eventual abandonment. Fig. 17 summarizes the system performance under different rejection rates.

As the rejection rate increases, the service rate decreases while the average waiting time increases. However, system profit does not necessarily decline. Since rejected drivers remain available for subsequent requests and no rejection penalties are imposed, the resulting supply shortage encourages the platform to increase both the price multiplier and the queue-jumping premium (see Fig. 17d). The higher revenue generated from each completed trip partially offsets the reduction in service volume. Although this mechanism helps sustain short-term profitability, it may reduce customer satisfaction and long-term demand, ultimately weakening the platform's competitiveness.

Overall, the proposed framework maintains stable performance under moderate driver rejection rates through adaptive pricing and dispatching decisions. Extending the framework to explicitly model driver incentives and compliance behavior constitutes an important direction for future research.

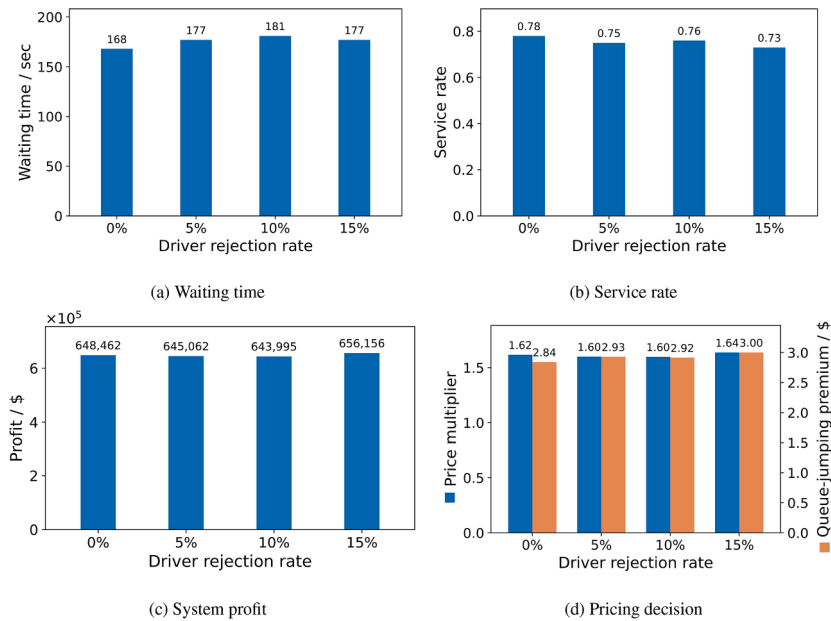


Fig. 17. Performance under different driver rejection rates.

5. Conclusion

This paper proposed an OD-based dynamic pricing framework that addresses key limitations in existing ride-sourcing strategies by incorporating heterogeneous spatial-temporal demand and customer-specific behaviors. The framework captured the diverse preferences of ride-sourcing users by incorporating customer-specific value of time (VOT) and willingness to pay (WTP) into an integral-form demand function. Customers with different VOT and WTP could choose to wait for a ride, abandon the trip, or jump the queue by paying a premium. The dynamic pricing problem is formulated as a constrained optimization problem that jointly determines origin-destination (OD)-based pricing and queue-jumping premiums. To effectively manage both short-term operational needs and long-term strategic goals, the framework incorporated a rolling horizon approach. This approach combined a short decision interval for real-time pricing adjustments with a longer planning horizon incorporating queuing dynamics and future travel demand. Furthermore, an event-triggered scheme was integrated to mitigate computational burden and stabilize pricing decisions, ensuring efficient and responsive price adjustments. Approximate Dynamic Programming (ADP) was employed to manage vehicle dispatching and relocation at the link-node level. By jointly optimizing OD-based pricing, queue-jumping premium, and link-node-level vehicle dispatching and relocation decisions, the proposed framework captures the complex interplay between demand elasticity, customer heterogeneity, and supply distribution. Empirical validation on the Manhattan road network confirmed the effectiveness of the proposed approach in improving platform profitability, increasing demand satisfaction, and enhancing overall system efficiency. These results suggest that incorporating behavioral and spatial-temporal heterogeneity into dynamic pricing and dispatching decisions is critical for building more adaptive and responsive ride-sourcing systems.

Future research could explore the trade-off between maximizing system profit and ensuring customer fairness within dynamic pricing strategies for ride-sourcing platforms. Priority-based matching may lead to longer waiting times for lower-paying customers, which could reduce customer loyalty and trust. Offering discounts or loyalty rewards to these customers may help mitigate fairness concerns while maintaining system efficiency. Another important direction is the design of incentive mechanisms to improve driver compliance, thereby enhancing the platform's ability to meet demand and improve service quality. Understanding how monetary incentives, reputation systems, or penalty structures influence driver behavior would strengthen the operational viability of the proposed framework. In addition, developing more accurate approaches to characterize heterogeneous demand distributions, particularly variations in customers' VOT and WTP, remains a critical avenue for improving model realism and decision effectiveness.

CRedit authorship contribution statement

Zheng Huang: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization; **Renxin Zhong:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization; **Tianlu Pan:** Writing – review & editing, Visualization, Formal analysis, Conceptualization; **Enming Liang:** Writing – review & editing, Visualization, Formal analysis, Conceptualization; **Agachai Sumalee:** Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization; **Yunping Huang:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization.

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Data availability

Data will be made available on request.

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